



Build an old people friendly PC

A step-by-step guide to transforming your existing PC to fit the needs of the elderly in your family.

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The elders of your house, at their special age, have special needs for most activities. Computing is no different. A computer always helps in staying connected and makes most tasks easier, especially for the elderly. But have you ever tried explaining to your grandmother what Facebook is, or how to use Skype? The learning curve hasn't been easy and specialised computers for them have mostly been too pricey. For example, the Telikin 18-inch model goes for 699\$, that is almost ₹50,000. Even the Senior Tablet for the elderly costs 400\$. You can achieve the same results by putting little effort into modifying your old personal computer, which would just be lying around anyway, into a completely usable and appropriately calibrated device for the seniors in your family. After all, they deserve to be connected too.

The external changes

You can modify the hardware in a number of ways to make the entire experience much more accessible for the elderly. There are some ways in which it is easier or more natural for them to interact with an electronic device, like using a remote, or directly touching where they want to click. Here are a few steps you can take without going overboard.

Step 1: Convert your monitor into a touchscreen

A touchscreen is much more intuitive to use compared to a keyboard and mouse inter-

face. Just look at how quickly anyone learns to use a smartphone. To make your monitor touch-sensitive, you will have to acquire a Digitiser, which is what senses the touch. You can get a DIY Touchscreen kit for your screen size online. The basic process to do this is to place the Digitiser screen on top of the screen inside the casing. You will need to remove the casing to do this and you may also need to make certain size adjustments. Connect a USB-to-4pin or any other



The very useful EldyTV app, which makes it easier for users to watch EldyTV's Italian channel along with YouTube, TED videos, Hulu etc.

specific adapter you may need between the touchscreen and the PC, and then load the required drivers. You're all set.

Step 2: Put in a Remote Control.

A remote control would greatly minimise the need to walk to the PC again and again just to change a song, or turn off the monitor. This can be done in a number of ways, both hardware and software-wise. A simple on-off switch can be installed by just replacing the connection between the power switch and the motherboard with a connector to an IR sensor. With some minor drilling and placement, the resulting arrangement would work with the power switch of most remotes. For a more complicated approach, you can use an application or code one yourself to detect all functions of a universal remote.

A much better and easier to use approach is using an App-Server based

approach in which you install a remote control app on a smartphone and install the server end tool for the same client on the PC. Unified Remote (<http://dgit.in/UniRem>) is one such app that supports more than 90 remotes for all kinds of applications with simple and easy to use buttons.

Step 3: Simplify the Cable Mess

Even the most tech-savvy ones among us are often daunted by the prospect of dealing with PC cables. One of the easiest ways to simplify this for the elderly is to use labels for everything. First, you will have to disconnect all the cables carefully, making sure not to open anything when you don't know what it is connected to. Once that is done, group the cables according to type and tie them together using Velcro strips – easy to open and tie again. For each such group, add a colour coded label, like 'Power', 'Display', 'Storage' etc.

Tweaking from within Windows

Windows offers a lot of in-built support features for the elderly which make the OS highly accessible. Most of them can be found in the Control Panel under the Accessibility Settings or Ease of Access centre under neatly grouped tabs/menus. Here are some we found useful.

Display Properties: Decrease the resolution of your display to make icons and menus larger. To retain sharpness and visibility, increase the dots per inch. Within the Settings tab in Display properties, click on Advanced > General > DPI setting > Large size (120DPI). Also use higher contrast. Go to Display Properties > Appearance. In the Windows and Buttons dropdown, select Windows Classic, then go for High Contrast in the Colour Scheme Menu. Also set the Font menu to Large. You can also individually tweak font details in the menu under the Advanced button,